

EXPERIMENTS AND MODELING OF THE HYDRAULIC RESISTANCE OF IN-LINE SQUARE PIN FIN HEAT SINKS WITH TOP BY-PASS FLOW

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ABSTRACT

In pin-fin heat sinks, the flow within the core exhibits separation and hence does not lend itself to simple analytical boundary layer or duct flow analysis of the wall friction. Indeed, form or bluff-body drag may be even more important than viscous wall friction in establishing the overall pressure drop. In this paper, we present some early findings from an experimental and analytical study aimed at obtaining physical insight into the behavior of square, in-line pin fin heat sinks. In the experimental study, various size aluminum heat sinks were utilized, where the pin heights were 12.5 mm, 17.5 mm, and 22.5 mm, and the base dimensions were kept fixed at 25 x 25 mm. A “two-branch by-pass model” was developed, in which a one-dimensional differential approach was used to model the fluid flow through the heat sink and its top by-pass duct. Central to the analysis is the introduction of accurate correlations for inlet, exit and core pressure drop within the heat sink. Pressure drop predictions correlated well with the experimental data when pressure drop coefficients measured for the heat sinks were used. Agreement was poorer when classical circular tube bundle pressure drop correlations were utilized, but the accuracy was high enough for design purposes. Inlet and exit pressure losses were as important as the core pressure drop in establishing the overall flow and pressure drop. Available literature models for inlet and exit effects were mostly adequate.

KEYWORDS

In-line, pin fins, heat sink, by-pass, heat sink, two-branch model, hydraulic resistance.

NOMENCLATURE

a fin side (m)
 A_{by} cross-sectional area of the by-pass channel (m²)
 A_{fhs} frontal area of the heat sink (m²)
 b distance between two fins in the flow direction (m)
 BR_{front} frontal by-pass ratio

BR^* reference by-pass ratio
 CL clearance ratio (L/H)
 d_f fin (hydraulic) diameter (m)
 d_h hydraulic diameter of the by-pass channel (m)
 f friction factor
 F_{by} friction forces in the by-pass section (N)
 F_{ff} friction forces in the finned section (N)
 H height of the finned section (fin height) (m)
 L height of the top by-pass section (top clearance) (m)
 P static pressure in the finned section (Pa)
 P_b static pressure in the by-pass section (Pa)
 P_{st} static pressure (Pa)
 P_{tot} total pressure (Pa)
 P_L non-dimensional longitudinal pitch (S_L/d_f)
 P_L^* reference non-dimensional longitudinal pitch
 P_T non-dimensional transverse pitch (S_T/d_f)
 Re_{max} Reynolds number based on maximum velocity in the finned section and the fin diameter
 Re_{app} approach Reynolds number- Reynolds number based on the approach velocity in the finned section and the fin diameter
 S_L longitudinal pitch (m)
 S_T transverse pitch (m)
 u velocity in the finned section (m/s)
 u_b velocity in the by-pass section (m/s)
 u_{app} approach velocity (m/s)
 u_{max} average maximum velocity between fins (m/s)
 DP_{tot} overall pressure drop in the finned section, Equation (8) (Pa)
 DP_{core} pressure drop in the finned section determined by the tube bundle correlations (Pa)
 DP_{ent} pressure drop in the finned section at the entrance (Pa)
 DP_{ex} pressure drop in the finned section at the exit (Pa)
 d uncertainty
 c correction factor for the tube bundle pressure drop
 m absolute viscosity (Pa · s)
 ρ density (kg/m³)

INTRODUCTION

The constant demand for speed and performance in electronics is unfortunately accompanied by ever increasing thermal dissipation. Air heat exchangers or “heat sinks” continue to be the most viable thermal solution for the electronics industry primarily because of low cost and high reliability. Heat sinks for electronics depend on conduction from the electronic package to heat sink base, followed by conduction into the extended surfaces and convection to the flow. A unique aspect of heat exchangers used in electronics is the possibility that the flow can bypass the heat exchanger core entirely, and leak from the core to the by-pass duct. Many authors have undertaken the characterization of parallel plate heat exchangers. Knight et al. [1] analytically characterized the flow and heat transfer behavior of such heat exchangers as function of geometry and fluid characteristics, for developing and fully developed flow. Teertstra et al. [2] similarly found analytical solutions for such configurations, and blended both regimes into a single solution for a wide range of Reynolds number. Lee [3] studied the effects of flow by-pass, by allowing flow around the heat exchanger in a partially confined flow configuration. Butterbaugh and Kang [4] treated this case by constructing a nodal network of flow paths, and detailed the calculation of each network element. They accounted not only for flow bypass, but also for that part of the flow that enters the heat sink and exists through the top, the so-called tip leakage.

For the case of pin fin configurations, however, work is sparse. The obvious reason for this is that within pin fin heat sinks, the flow is dominated by flow separation and complex three-dimensional flow at the pin-base junction. Hence, simple closed form analytical models of the friction and heat transfer coefficients are not possible. By necessity models thus are simple one-dimensional formulations with empirical formulations of friction and heat transfer coefficients [5, 6] or fully conjugate 3-D CFD solutions [7, 8, 9].

Except for the pin-end wall effects and tip leakage, the flow within the pin-fin heat sink resembles that within an array of tubes of infinite length. Arrays of circular tubes have been extensively studied [10, 11, 12] but detailed experimental data on arrays of tubes of square or rectangular cross-section are rare, except for rectangular cross-sections of large aspect ratio used in in-line and off-set strip fins [6, 10, 11].

Kays and London [10] present data for infinitely long round tubes in cross flow. Idelchik [11] presents correlations for several different geometries of bundles of tube. Zhukauskas [12] also presents experimental correlations for tubes in cross flow, as a function of Reynolds number, pin diameter and pitch, with correction factors for longitudinal pitch different to tangential pitch. Data of this type, although not specified for such small heat exchangers as those within the scope of this paper, have traditionally been used by designers, due to the lack of a better methodology.

For such small heat exchangers, Shaukatullah et al. [5] measured the thermal performance for in-line square pin fins and plate heat sinks for different fin thickness, spacing, height, and angle of approach for velocities under 5m/s, while allowing flow to partially bypass the exchanger. Jonsson and Moshfegh [6] also attempted to characterize plate and circular, rectangular and strip pin fins, in both staggered and in-line configurations for different dimensions, allowing for variations on tip and side bypass. Computational fluid dynamics (CFD) approaches have been extensively applied to study flow in heat sinks, for example by Jonsson and Moshfegh [7], Biber and Belady [8] and Dvinsky et al. [9].

Dvinsky et al. [9] used CFD to study in detail flow and heat transfer behavior for in-line and staggered heat sinks for 1, 3 and 5m/s. By allowing top and side flow bypass, it brings to our attention the deficiencies of a simple 2- (or even 4-) path model that does not allow for leakage from the heat sink, back to the channel. They found that, when allowing side and top clearance, up to half of the flow that enters the heat sink may leave it through the side and top. CFD simulations of Jonsson and Moshfegh [7] revealed that this leakage of flow may be quite complex. They reported inward flow from the sides, to balance flow lost through the top.

Motivation

Careful experimental characterization of square pin fin heat sinks is lacking in the body of literature pertaining to air-cooled heat exchangers or heat sinks used in electronic systems. This work is therefore motivated by a general lack of data that can be used for heat sink optimization and design and simulation of their behavior. This paper has three goals:

- To present carefully documented experimental data on the hydraulic behavior of a consistent set of square pin fin heat sinks of variable height and pitch.
- To compare the hydraulic resistance of square pin fin heat sinks of finite length to the resistance of circular tube arrays of infinite (or very long) length available in the literature.
- To evaluate the accuracy of a simple two-branch by-pass model in predicting overall pressure drop, in particular by comparing frictional coefficients for arrays of circular tubes from the literature, and those measured for these heat sinks.

EXPERIMENTS

Experimental Apparatus

The test geometry is described in Figures 1a and 1b. The test apparatus was designed to allow tip by-pass variations but not side by-pass. A duct of square cross-section was constructed with 0.5inch Plexiglas walls, where the sides and the floor are fixed and the top is movable. The test section runs at sub-ambient pressure, as a way to better control flow and velocity conditions. A quarter-circle cross-section flare was placed at the mouth of the channel, as a smooth contraction from the ambient, into the channel. A Pitot tube placed 5.1cm downstream from the entrance (25cm upstream from the heat sink) measured the total pressure, which was compared to the

first static pressure tap, 18cm from the entrance, to measure approach velocity, u_{app} . During testing, the heat sink is placed on the top, movable wall, which is adjusted to allow for tip bypass, according to the test case. This movable wall is equipped with 18 pressure taps distributed as shown in Figure 2, on the centerline. The upstream face of the heat sink is located 30cm from the inlet. The heat sink rests on a seat, such that the wall is flush with the base of the heat sink, and only the pins protrude. Downstream from the channel, a honeycombed inlet plenum has a bypass gate to ambient, used to adjust the flow velocity within the test section. Flow is induced through the channel by a squirrel cage blower, which is attached to the exit of the plenum.

The heat sinks used in the study are a subset of the set used by Shaukatullah et al. [5] and are summarized in Table 1. The heat sinks are machined aluminum. All of the heat sinks have the base dimensions of 25mm x 25mm x 2.5mm. Pin heights range from 12.5mm to 22.5mm. Experiments were conducted on 4x4, 5x5, 6x6, 7x7 and 8x8 pin arrays, for $a=1.5$ mm, and 5x5 arrangement with $a=2$ mm and 2.5mm.

SLA Test Apparatus

In this paper, we will primarily report on overall pressure drop from the inlet to the exit of the aluminum heat sinks. To evaluate a novel method for measuring pressure distribution within the heat sink, a stereolithography (SLA from here forward) model of heat sink 7C was designed with static pressure taps on the base and on the pins, as shown in Figure 4. All pin pressure taps have the same orientation. Because of the success of the SLA model, measurements are underway on SLA models of the entire set.

Uncertainty

Static pressure was measured with the apparatus shown in Figures 2 and 3. A Baratron model 698A differential pressure transducer with a model 670 signal conditioning unit outputs a voltage that is proportional to the local static pressure. This voltage is a real time signal of the pressure, and is averaged over 3ec (4sec for the SLA model, sampling frequency of 1KHz), with a National Instruments AT-MIO-16A-2 A/D converter. The set-up also includes a scanning valve that allows toggling between the different taps. To account for transient effects, the data was not collected until after the valve was parked for three seconds in the same tap.

Table 2 shows the different sources of uncertainty, a brief description and its value. Using this information, for each static pressure measurement, P_{st} , the uncertainty will be:

$$dP_{st}^2 = 0.0066^2 + (0.0008 P_{st})^2 \quad (1)$$

and the uncertainty for the static pressure drop DP_{st} will be:

$$dDP_{st}^2 = dP_{st-up}^2 + dP_{st-down}^2 \quad (2)$$

where P_{st-up} and $P_{st-down}$ are the upstream and downstream static pressures, respectively. Typically, the percent uncertainty calculated for pressure dP_{st}/P_{st} is in the order of 0.1%, and dDP_{st} to DP_{st} ratio is in the order of 0.2%. As in any internal flow testing, the flow unsteadiness induced fluctuations in the instantaneous wall pressure. Time histories show the RMS of the fluctuations to be on the order of 1% of the mean.

Experimental Results

Pressure data were collected for all heat sinks for $CL=0.00, 0.25, 0.50, 1.00, 2.00$, and 3.00 with approach velocities of, $u_{app}=2$ m/s and 4 m/s. Additional tests were performed for $CL=4$ with the heat sinks of $H=12.5$ mm and 17.5 mm at the same approach velocities.

For all cases, the total pressure drop, DP_{tot} , was determined by using the lowest pressure upstream of the heat sink, and the highest pressure downstream from the heat sink, in other words, it was extracted from the difference between P_{up} and P_{down} .

Figures 5 and 6 show the pressure profiles of the channel for heat sinks 1C and 4C for $u_{app}=4$ m/s. These two cases correspond to the tallest, densest, and most sparse heat sinks, for $a=1.5$ mm. As expected, the pressure drop decreased as CL and b increase, with diminishing effects from pin spacing at higher clearances. This behavior is consistent with the set, and can also be appreciated in Figure 7. It should be noted that the pressure increases on the pressure tap directly in front of the heat sink. This profile is thought to be due to stagnation as the flow encounters the obstruction. The acceleration at the inlet of the heat sink leads to an inlet pressure drop, however this could not be observed in the data, as there were no pressure taps in the heat sink interior. The pressure increase and recovery are more dramatic in 1C compared to those of 4C, possibly due to the local position of the tap in question. The taps (on the spanwise center line) lie in front of or behind a pin and this will be the case for all heat sinks with an odd number of pins per row, but in between pins for heat sinks with an even number of pins per row. Interpretation of these local effects should be done with care.

Figure 7 shows the effects of pitch size on total pressure drop vs. top clearance (CL) for all of the heat sinks of $H=22.5$ mm tested with $u_{app}=4$ m/s. It can be seen that the effect of heat sink geometry diminishes as the bypass increases. At $CL=0$ the DP_{tot} span is 22Pa, while at $CL=3$ the DP_{tot} span is only 5Pa. The upper bound shown in Figure 7 is for a solid aluminum block of height $H=22.5$ mm which represents a heat sink with $P_L=1$. It can be noted that the non-dimensional transverse pitch alone does not seem to fully capture the geometrical effects on the flow behavior in such short pin fin heat sinks. For a constant pin diameter, pressure drop monotonically increases with decreasing pitch. However, changing the pin diameter alters this behavior considerably. This may be due to a change in the number of rows. For instance, while heat sink 7C has almost the same non-dimensional pitch as 1C, it has only 5 rows, while 1C has 8. Likewise, 6C (5 rows) behaves more closely to 4C (5 rows) than 3C (6 rows) and 2C (7 rows).

The experimental data demonstrated inconsistent behavior for $CL=0.25$ and 0.50 for both $u_{app}=2$ m/s and 4 m/s. The trend of increasing DP_{tot} with decreasing CL was reversed, or DP_{tot} was measured to be higher than for the $CL=0$ case, for a given heat sink. At low clearances, pressure measurements did not exhibit high repeatability. Discrepancies up to 10% were found between identical tests. Data for $CL=0.25$ and 0.50 reveal interesting trends, but should be viewed as tentative.

Figure 8 shows DP_{tot} vs. P_L data for $u_{app}=2$ and 4m/s , and $CL=0$. It illustrates pressure drop dependence on fin spacing and diameter. Again, this curve will reach infinity in the case of $P_L=1$, and its minimum value in the case of $P_L \rightarrow \infty$.

Figure 9 shows preliminary static pressure measurements made within the core of the heat sink 7C as illustrated in Figure 4. The positions of the static taps follow the labeling shown in Figure 1a. The heat sink was fabricated using the SLA rapid prototyping technique, which made it possible to fabricate numerous static pressure ports rather easily. Discrepancy was observed in DP_{tot} measure for the SLA prototype, and the aluminum version of the same heat sink. This is believed to be a result of small differences in the dimensions due to shrinkage of SLA model, as well as differences in surface finish. Nevertheless, these local core pressure measurements clearly show the inlet pressure drop, and the exit pressure recovery.

The dependence of overall pressure drop on approach Reynolds number is shown in Figure 10. For a given d_f and Re_{app} , overall pressure drop increases with a decreasing non-dimensional longitudinal pitch, as expected. However, keeping the non-dimensional pitch size fixed as the number of fins and fin diameters are changed at a given approach velocity, there is a significant difference between two data sets corresponding to two different fin diameters. Hence, as mentioned previously, it should be noted that there is clearly a separate dependence on pin diameter.

COMPUTATIONAL ANALYSIS

Governing Equations

Given the mass flow rate and the total pressure of the air at the inlet of the heat sink in addition to the geometric parameters, pressure drop can be determined on a differential basis by utilizing a simple one-dimensional model, which incorporates empirical friction factor relations. Most generally, a duct can be composed of two different sections, namely a finned section (heat sink) and a by-pass channel at the top of it as shown in Figure 12. The “two-branch by-pass model” [4] assumes that mass flow rate entering either section does not change along the flow direction, in other words, tip leakage from the finned section to the by-pass channel is neglected and therefore, mass flow rate remains constant in each section.

To develop a simple one-dimensional flow or hydraulic model, the velocities in either section are assumed to be uniform. Hence, for the differential control volumes of the finned (CV-1) and the by-pass section (CV-2) shown in Figure 12, the momentum equations in the flow direction can be expressed for the finned section as,

$$-(r u_{x+dx} A_{fhs}) u_{x+dx} + (r u_x A_{fhs}) u_x = P_x A_{fhs} - P_{x+dx} A_{fhs} - dF_{ff} \quad (3)$$

and for the by-pass section as,

$$-(r u_{b,x+dx} A_{by}) u_{b,x+dx} + (r u_{b,x} A_{by}) u_{b,x} = P_{b,x} A_{by} - P_{b,x+dx} A_{by} - dF_{by} \quad (4)$$

where dF_{ff} and dF_{by} represent the frictional forces exerted on the given control volume of heat sink and by-pass duct, respectively. Since the tip leakage flow is neglected, the terms of the LHS cancel in both equations, and therefore vanish. If the outlet pressure terms are expressed by their first order Taylor series, Equations (3) and (4) can be rearranged and written in a more compact form,

$$A_{fhs} dP = -dF_{ff} \quad (5)$$

$$A_{by} dP_b = -dF_{by} \quad (6)$$

Equations (5) and (6) simply state that for steady flow, the net pressure force is balanced by the net drag force. Since the drag for complex flows cannot be expressed analytically, it is customary to introduce an empirical drag or friction factor defined in a form suggested for example by Zhukauskas [12],

$$DP = f \left(\frac{r u^2}{2} \right) c \quad (7)$$

where f is the Darcy friction factor, r is the density of the fluid, u is the reference velocity in the channel and c is a correction factor. For the finned section, f can be determined for the Reynolds number, Re_{max} , based on the hydraulic diameter of the tube or the fin, the average maximum velocity between the fins, the geometric parameters and the type of the fin arrangement. c depends on the finned section dimensions and the type of the fin arrangement only. For the by-pass duct, Re (based on the hydraulic diameter of the channel) and the duct average velocity are sufficient to determine f . c is equal to 1 for the by-pass duct.

The relation given by (7) is valid for determining the pressure drop in tube bundles or in channels, however, the total pressure drop experienced in a heat sink should also include the entrance and the exit effects of the heat sink. This situation is illustrated in Figure 1a and the total pressure drop for the heat sink can be written as,

$$DP_{tot} = DP_{ent} + DP_{core} - DP_{ex} \quad (8)$$

where ΔP_{ent} is the pressure drop at the entrance of the heat sink, ΔP_{core} is the pressure drop due to the fins, and ΔP_{ex} is the pressure rise at the heat sink exit. Referring to Figure 1a, they can also be expressed in the form such as,

$$DP_{ent} = P_{in} - P_B \quad (8a)$$

$$DP_{core} = P_B - P_F \quad (8b)$$

$$DP_{ex} = P_G - P_F \quad (8c)$$

The terms, ΔP_{ent} and ΔP_{ex} are shown separately in Equation (8), since they comprise both frictional effects and inviscid effects due to the geometry change, i.e. contraction and expansion at the entrance and expansion at the exit. ΔP_{ent} and ΔP_{ex} can be expressed as [10],

$$DP_{ent} = \frac{r u_{max}^2}{2} (1 - s^2 + k_{cont}) - \frac{r u_{max}^2}{2} (1 - s^2 + k_{exp}) \quad (9)$$

$$DP_{ex} = \frac{r u_{max}^2}{2} (1 - s^2 - k_{exp}) \quad (10)$$

where s is the ratio of the frontal area to the free-flow area, $u_{\max}$ is the average maximum velocity between the fins, and k_{cont} , k_{exp} are the friction coefficients to account for viscous effects at the entrance and at the exit, respectively. For the comparisons made in this paper, k_{cont} and k_{ex} are taken from Figure 5.3 of [10] for a simple, two-dimensional rectangular contraction or expansion. It should be noted that the entrance pressure drop includes a contraction into the first pin row followed by an expansion out of the pin row.

Methodology

Pressure drop for the heat sinks with no by-pass can simply be determined by using the developed one-dimensional model marching in the flow direction. However, with by-pass, prediction of the pressure drop in the heat sink requires an iterative procedure as shown in the proposed computational algorithm illustrated in Figure 11. This is due to the fact that the static pressures at the exit of the heat sink and its by-pass duct should be equal to each other and the pressure drops are different for the two sections. If there is a by-pass channel, the model first assumes that the inlet velocity of the air for the two sections is same, then by decreasing the velocity of the air in the finned section and therefore, by increasing the velocity of the air in the by-pass duct with a corresponding increment according to the conservation of mass requirements, a convergent solution is sought. Increasing the velocity causes a decrease in the static pressure or vice versa, provided that the total pressure at the inlet is fixed. The model calculates the exit static pressures for both of the sections and determines whether they are equal or not. If the difference in pressures is less than 0.5% (specified convergence criteria) with respect to the pressure drop of the finned section, then they are assumed to be equal to each other. This procedure is repeated until the equal exit static pressures are obtained.

RESULTS AND DISCUSSION

In order to compare the experimental results with those in literature, empirical friction factors for the core loss were extracted from the data taken with no by-pass. Figure 13 shows the experimental core friction factor data with respect to the non-dimensional pitch in the case of no clearance. The procedure to extract the friction factor from the experimental data can be explained as follows: First, since the ΔP_{ent} and ΔP_{ex} were not directly measured, they are determined by Equations (9) and (10), respectively, for a known mass flow rate of the air. Second, the pressure drop inside the heat sink, ΔP_{core} , is determined by deducting the entrance and exit pressure drops calculated with Equations (9) and (10) from the experimentally determined overall pressure drop from Equation (8). Third, with the known values of ΔP_{core} and c , f per row of fins can be calculated by using Equation (7).

The results are presented for two different Re_{app} of 179 and 358. As it can be seen from the figure, f decreases with non-dimensional pitch up to $P_L \cong 2.65$ and increases thereafter. Qualitatively, this agrees well with aligned arrangement tube bundles for $P_L < 3$, in which there is a sudden decrease in

pressure drag of the in-line tubes due to the influence of the previous rows [12].

In order to compare finite length and infinitely long tubes in terms of friction factor, f is plotted as a function of Re_{max} in Figure 14. A direct comparison is not possible, since the non-dimensional pitch in the heat sinks are not the same as those in [12]. In contrast to the results shown in [12], f increases with increasing Re_{max} . It is not reasonable to expect the same behavior. We believe that the discrepancy occurs due to the end-wall effects, and to the possible inadequacy of the entrance and the exit effects predicted by Equations (9) and (10), respectively. It is worth noting that the factors stated above are neglected in determining the friction factor of infinitely long tube bundles [12].

In Figure 9, a comparison is made between the measured static pressure in the SLA prototype of heat sink 7C and the predicted pressure using the one-dimensional hydraulic model, utilizing the core friction coefficients from Zhukauskas [12] for arrays of infinitely long circular tubes, and the inlet and exit loss coefficients from Kays and London [10]. For this single test case, the computational model predicts the pressure behavior well. There is a good agreement for all of the separate sections, i.e. the inlet and the core pressure drop and the exit recovery. It is also noteworthy that the inlet pressure drop is as significant as the core pressure drop in the overall flow resistance. On the other hand, the difference between the core pressure drops is due to the end-wall and entrance region effects, i.e. correlations in [12] were developed for long bank of tubes, in which most of the flow was in the fully developed region, however that might not be the case for the short heat sink used in the experiments.

Figures 15 and 16 demonstrate the overall pressure drop at various clearance ratios for two different non-dimensional pitches of 2.238 and 3.917, which correspond to the densest and most sparse in-line fin arrangements, respectively. The figures exhibit that at high clearance ratios, there is a good agreement between the theoretical predictions and the experimental results. However, the discrepancy between the results increases as the clearance ratio decreases. This discrepancy, once again may be attributed to the factors as mentioned previously for f .

Development of a New By-pass Model

With the reasonably good validation of the model using friction coefficients for circular tubes, the model can be used to further expand our understanding of the hydraulic behavior. In other words, the results can be presented in a non-dimensional and compact form by introducing a new by-pass model. Using the computational model, Figure 17 shows that the frontal by-pass ratio, BR_{front} decreases with increasing the non-dimensional pitch at clearance ratios of 0.00, 0.50, 1.00 and 2.00 at $Re_{app}=358$. BR_{front} is defined as,

$$BR_{front} = \frac{u_{app} - u}{u_{app}} \quad (11)$$

and represents the percentage of the flow that bypasses the heat sink to the flow that approaches the heat sink in its projected frontal area, as illustrated in Figure 1a (It should be noted that this is equivalent to the by-pass ratio defined in [7]).

Figure 18 demonstrates that BR_{front} increases with increasing CL , as expected. A careful observation of Figure 17 together with Figure 18 reveals that the product of the by-pass ratio and the non-dimensional pitch yields a constant value for a given CL and a Re_{app} . Normalizing the by-pass ratios with respect to a chosen reference, the normalized by-pass ratio can be defined as,

$$BR_{front} = BR^* \left(\frac{P_L^*}{P_L} \right) \quad (12)$$

where BR^* is the reference by-pass ratio and P_L^* is the corresponding reference non-dimensional longitudinal pitch, taken as $P_L^* = 3.133$ for the 6*6 square arrangement.

It should be mentioned that the P_L^{-1} dependence in Equation (12) deviates considerably for $CL > 3.00$. On the other hand, computations at Re_{app} of 179, 268, 358 and 447 demonstrate that the Re_{app} dependence is weak and can be neglected for $0.00 \leq CL \leq 3.00$.

The results are plotted as $(1-BR^*)$ vs. CL in Figure 19 at $Re_{app}=447$. Two separate regions can be identified from Figure 18. The first region is for $0.00 \leq CL \leq 0.25$, where the quantity $(1-BR^*)$ decreases rapidly with clearance ratio. The second region is for $0.25 < CL \leq 3.00$, where the decrease in $(1-BR^*)$ becomes considerably smaller with increasing clearance ratio. The first region shows an exponential dependence of $(1-BR^*)$ against clearance ratio, whereas, the second region exhibits a negative power dependence. This can be expressed as a piecewise function of clearance ratio by applying curve fits to each region, resulting in:

$$(1-BR^*) = \begin{cases} 0.996 \exp(-0.707CL) & 0.00 \leq CL \leq 0.25 \quad (13a) \\ 0.681CL^{-0.148} & 0.25 < CL \leq 3.00 \quad (13b) \end{cases}$$

Equations (13a) and (13b) provide a simple method for estimating the flow through the heat sink, which is useful in determining the amount of heat transferred.

CONCLUSIONS

Friction factor, pressure drop, and by-pass ratio results were demonstrated for in-line pin square pin fin heat sinks with various pitch sizes. A computational tool was constructed and the theoretical results extracted from the tool were compared with those of experiments. In order to make detailed pressure measurements, a SLA model was built and tested.

The following conclusions are drawn for the presented study:

1. In determining the hydraulic resistance of heat sinks, infinitely long tube bundle correlations can be used

with a reasonable accuracy, however end-wall, entrance and exit effects may result in a considerable difference between theoretical predictions and experimental results.

2. Comparison of SLA measurements with theoretical predictions shows that the estimation of entrance and exit effects is important; poor estimation of those may lead to significant differences in determining the overall pressure drop.
3. The product of by-pass ratio and the non-dimensional pitch yields a constant value, regardless of the number of fins in the heat sink. Therefore by-pass ratios can be determined based on a reference heat sink and shown by a single piecewise function with respect to clearance if the Re_{app} dependence is weak.

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Table 1 Geometric description of heat sinks tested.
n: number of rows in heat sink, all dimensions in mm.

Description	a	b	H	n	PL=(a+b)/a
1A	1.50	1.86	12.50	8	2.238
1B	1.50	1.86	17.50	8	2.238
1C	1.50	1.86	22.50	8	2.238
2A	1.50	2.42	12.50	7	2.611
2B	1.50	2.42	17.50	7	2.611
2C	1.50	2.42	22.50	7	2.611
3A	1.50	3.20	12.50	6	3.133
3B	1.50	3.20	17.50	6	3.133
3C	1.50	3.20	22.50	6	3.133
4A	1.50	4.38	12.50	5	3.917
4B	1.50	4.38	17.50	5	3.917
4C	1.50	4.38	22.50	5	3.917
5A	1.50	6.33	12.50	4	5.222
5B	1.50	6.33	17.50	4	5.222
6A	2.00	3.75	12.50	5	2.875
6B	2.00	3.75	17.50	5	2.875
6C	2.00	3.75	22.50	5	2.875
7A	2.50	3.13	12.50	5	2.253
7B	2.50	3.13	17.50	5	2.253
7C	2.50	3.13	22.50	5	2.253

Table 2 Sources and description of uncertainty.

Source	Value	Description
A/D resolution	+/-0.0065Pa	12bit, +/-10V span
Signal cond. accuracy	+/-0.008Pa (for 10Pa)	0.08% of reading
Pressure trans. accuracy	0.00133Pa	100ppm of FS*
*Full scale=.1mmHg=13.33Pa		

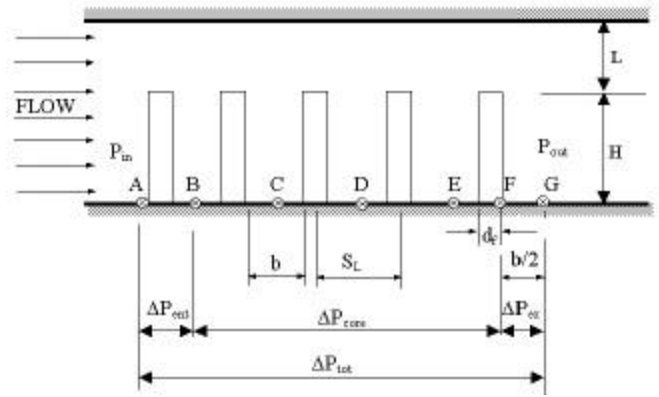


Figure 1a Illustration of a heat sink with a top by-pass channel (side-view).

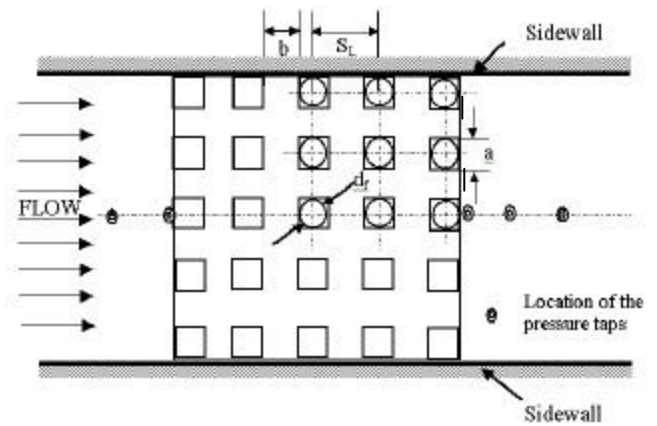


Figure 1b Illustration of a sample heat sink geometry.

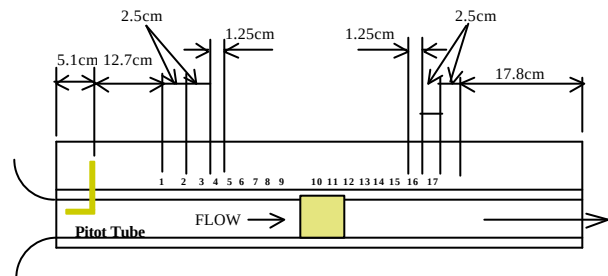


Figure 2 Schematic of the test section.

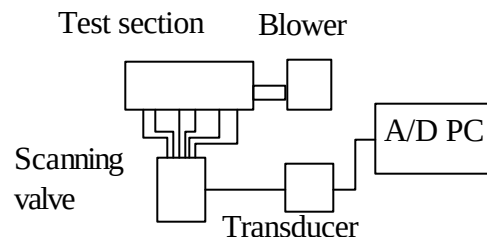


Figure 3 Schematic of the experimental set-up.

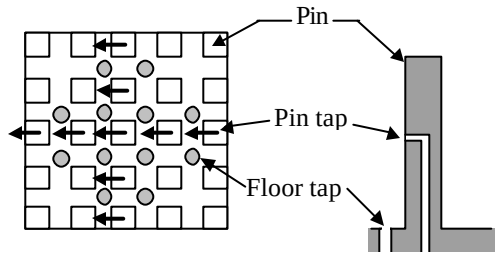


Figure 4 SLA model of heat sink 7C with two different pressure tap locations

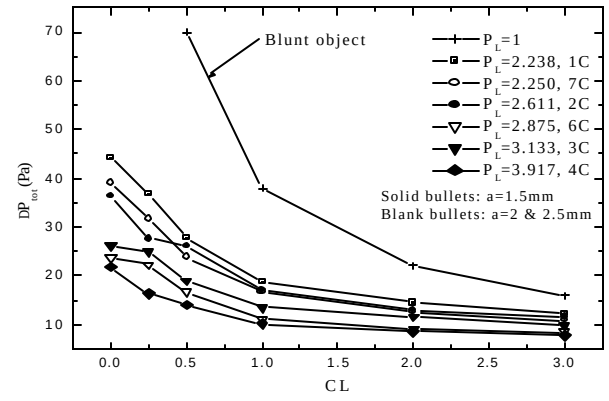


Figure 7 Total pressure drop vs. clearance ratio for $H=22.5\text{mm}$, $u_{app}=4\text{m/s}$ at various non-dimensional pitches.

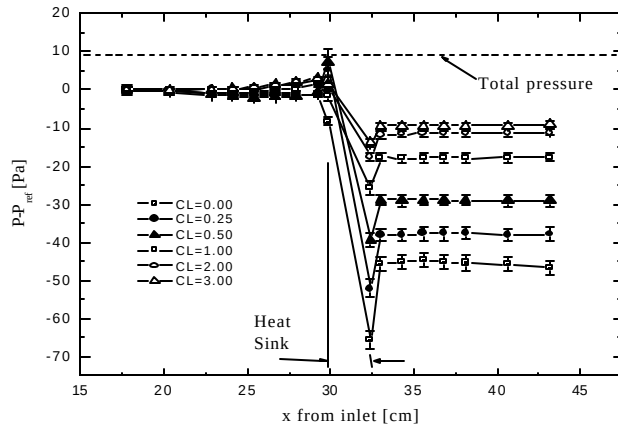


Figure 5 Channel pressure distribution for 1C, at $u_{app}=4\text{m/s}$ at various clearance ratios

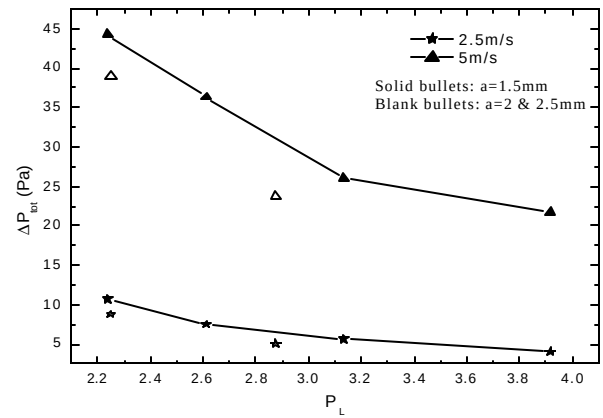


Figure 8 Pressure drop vs. non-dimensional pitch for $a=1.5, 2, 2.5\text{mm}$ and $H=22.5\text{mm}$ at $CL=0.0$.

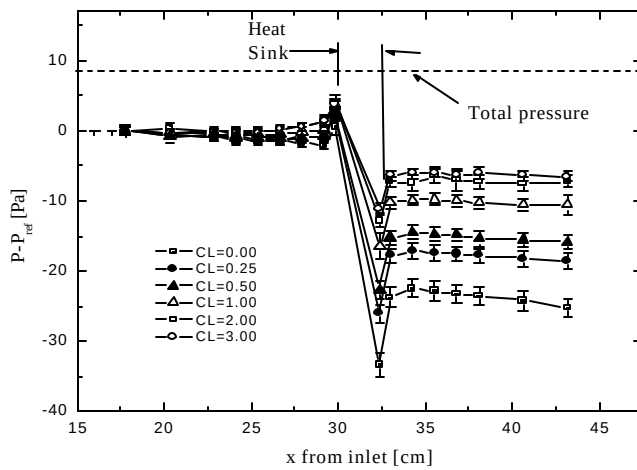


Figure 6 Channel pressure distribution for 4C, at $u_{app}=4\text{m/s}$ at various clearance ratios

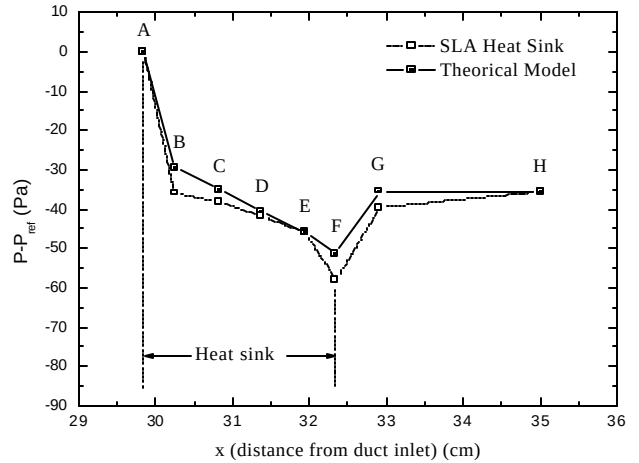


Figure 9 Comparison of the experimental pressure distribution inside the heat sink (SLA model) with the theoretical pressure distribution, for heat sink 7C, at $CL=0$ and $u_{app}=4\text{m/s}$.

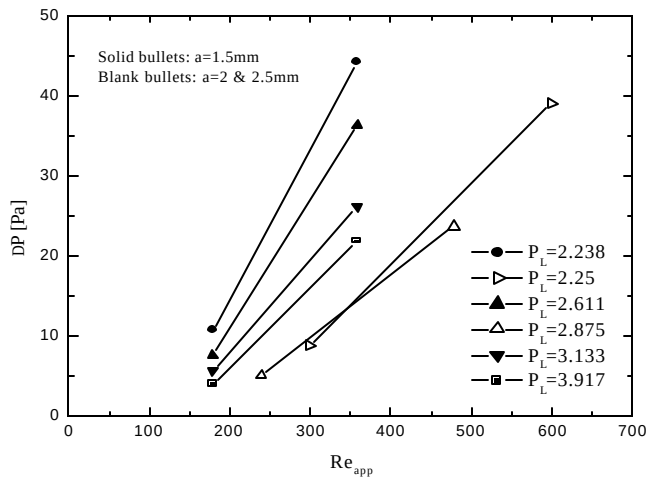


Figure 10 Experimental pressure drop vs. Re_{app} at various non-dimensional pitches.

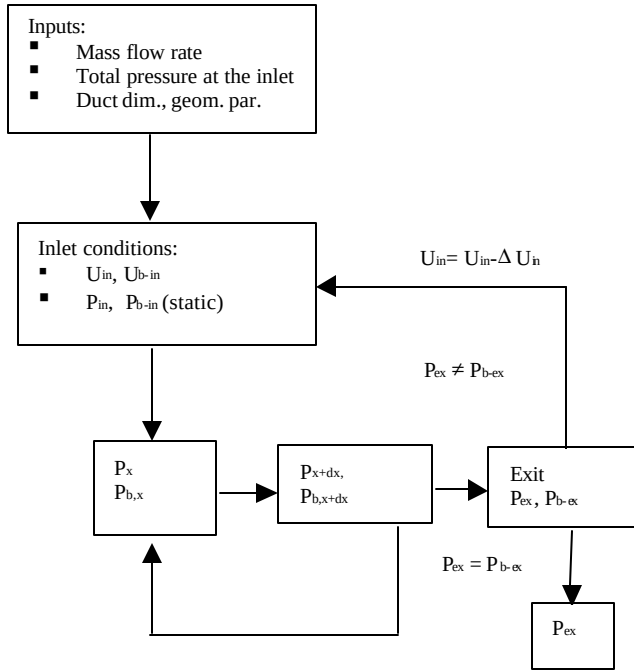


Figure 11 Block diagram demonstrating the computational procedure of determining the pressure drop in heat sinks.

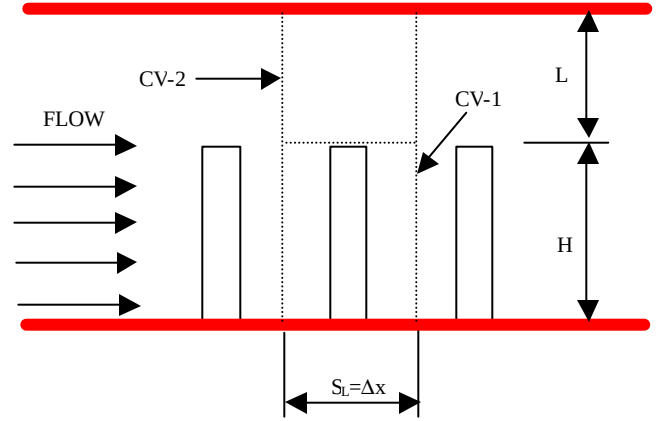


Figure 12 Control volumes for finned section and by-pass duct in the theoretical model.

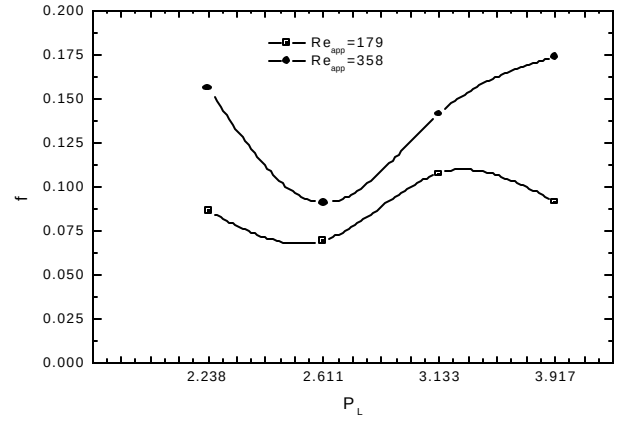


Figure 13 Measured friction factor vs. non-dimensional pitch at Re_{app} of 179 and 358.

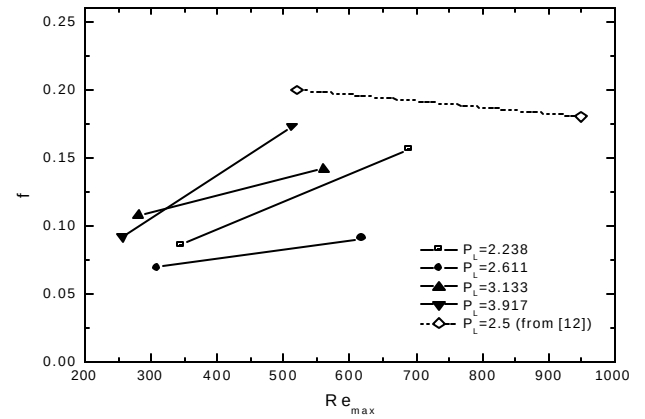


Figure 14 Measured friction factor vs. Re_{max} at various non-dimensional pitches.

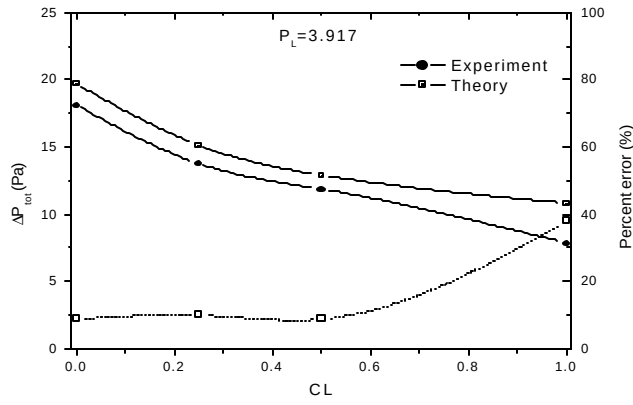


Figure 15 Comparison of measured and predicted total pressure drop (ΔP_{tot}) vs. CL at $P_L=3.917$ and $u_{app}=4\text{m/s}$. The bottom curve represents the percent error.

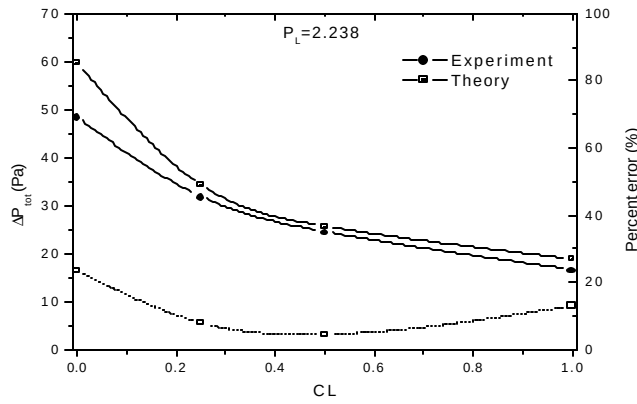


Figure 16 Comparison of measured and predicted total pressure drop (ΔP_{tot}) vs. CL at $P_L=2.238$ and $u_{app}=4\text{m/s}$. The bottom curve represents the percent error.

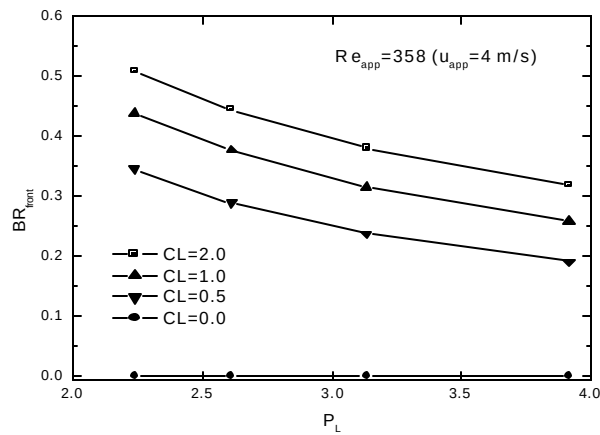


Figure 17 Frontal by-pass ratio vs. non-dimensional pitch at $Re_{app}=358$ and various clearance ratios.

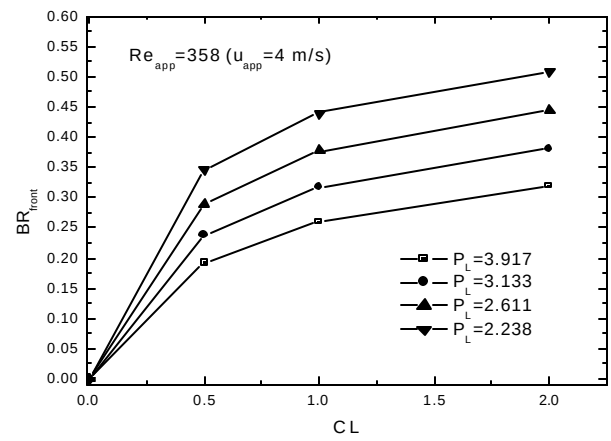


Figure 18 Frontal by-pass ratio vs. CL at $Re_{app}=358$ and various non-dimensional pitches.

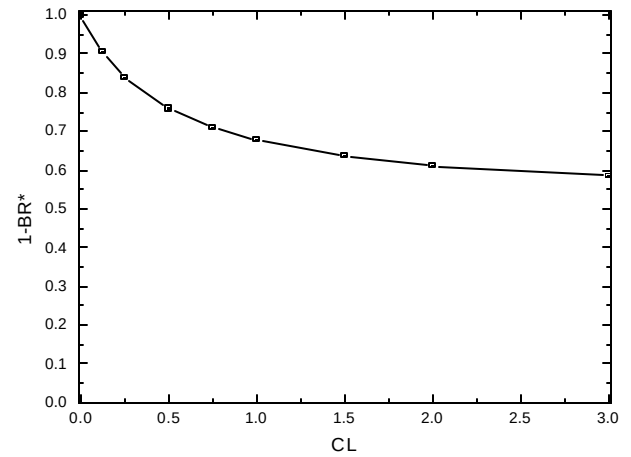


Figure 19 Quantity $(1-BR^*)$ vs. clearance ratio ($Re_{app}=447$).